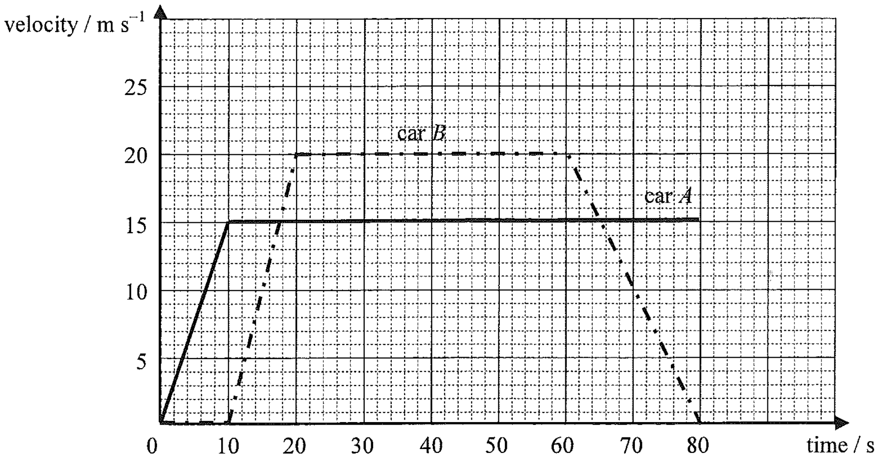
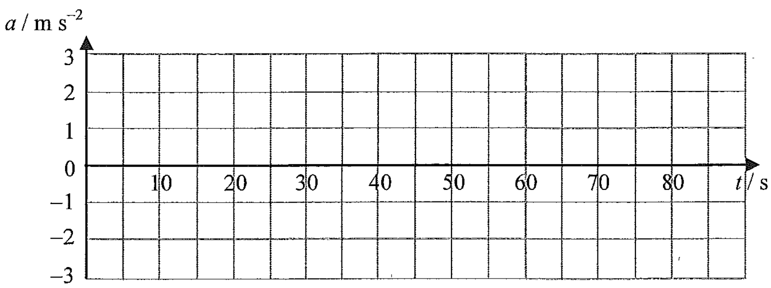
3. Two cars $A$ and $B$ initially at the same position, start to travel along the same straight horizontal road. The graph below shows how their velocities vary with time.




(a) Describe the motion of car $A$ along the whole journey from $t = 0$ to $t = 80$ s. (2 marks)

(b) (i) Which car attained the greatest acceleration throughout the journey ? Find this acceleration. (2 marks)

(ii) Sketch the acceleration-time ($a$ - $t$) graph of car $B$ from $t = 0$ to $t = 80$ s. (2 marks)

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(c) (i) At $t = 20$ s, what is the separation between cars $A$ and $B$ ? (2 marks)

(ii) Deduce the time at which car $B$ catches up with car $A$. (2 marks)

(d) Both cars are of similar size and shape. It is known that the total resistive force experienced by each car is proportional to the square of its velocity. Determine the ratio of power output of the engine of car $A$ to that of car $B$ within the period $t = 20$ s to $t = 60$ s. (2 marks)

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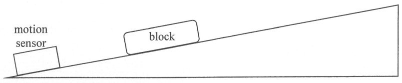
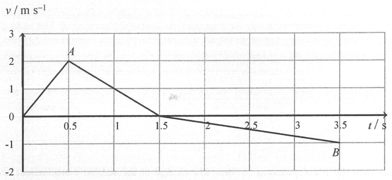
4. The motion of a block on an inclined plane can be investigated using a motion sensor connected to a computer (not shown in Figure 4.1).

Figure 4.1




A block is given a push up a rough inclined plane and then released. The velocity-time ($v$-$t$) graph recorded by the sensor is shown below. Assume that the frictional force acting on the block is constant in magnitude throughout the motion. Neglect air resistance. ($g = 9.81\ \mathrm{m\ s^{-2}}$)




Point $A$ on the graph corresponds to the instant at which the push is removed.

(a) Describe the block's motion **from $A$ to $B$**. (2 marks)

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(b) (i) Find the magnitude of the block's acceleration from $t = 1.5$ s to $t = 3.5$ s. (2 marks)

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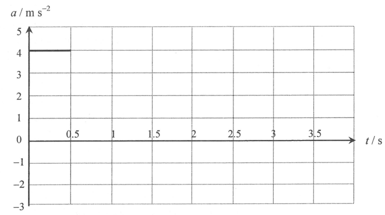
(ii) Draw the corresponding acceleration-time ($a$-$t$) graph of the block. With the direction up the inclined plane taken to be positive, the part during which the block is being pushed has been drawn for you. (2 marks)




(c) Draw a free-body diagram to show the force(s) (with labels) acting on the block as it moves up the inclined plane after the push is removed. (2 marks)

(d) If the mass of the block is 1.0 kg, find the magnitude of the frictional force. (3 marks)

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(a) Calculate the acceleration of the lift from $t = 0$ to $t = 2$ s. (2 marks)

The reading of the balance changes during the person's ride on the lift and the readings registered are 685 N, 569 N and 395 N.

(b) Match these readings with the three stages, $A$, $B$ and $C$, of the ride (shown in Figure 3.1). Hence deduce the mass of the person. (3 marks)

$A$:______ $B$:______ $C$: ______

(ii) Hence estimate the height of the building. (2 marks)







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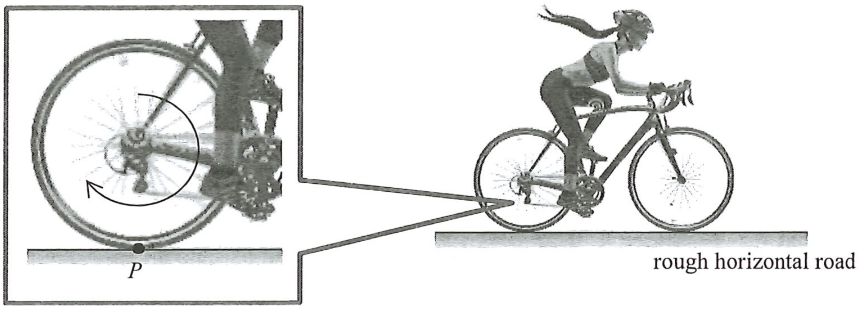
3. Figure 3.1 shows a cyclist riding on a bicycle travelling with a constant velocity of 8.0 m s$^{-1}$ on a rough horizontal road.

Figure 3.1

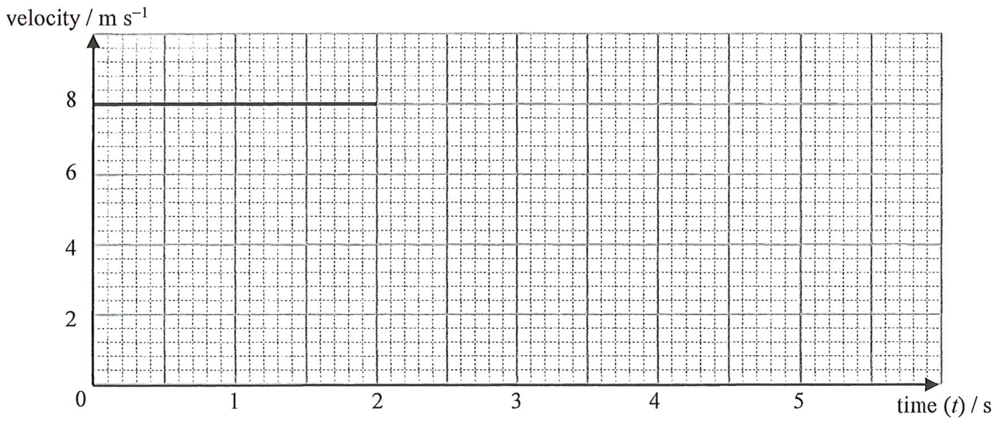



(a) The rear wheel is rotating clockwise without slipping on the ground as shown and $P$ is the contact point with the road at that moment. Indicate in the figure the frictional force acting on the rear wheel at $P$. (1 mark)

(b) If the effective resistive force experienced by the cyclist and the bicycle is 17.0 N when riding at this constant velocity, estimate the mechanical power delivered by the cyclist. (2 marks)

(c) At time $t = 2.0$ s, the cyclist sees a small obstacle 9 m directly in front of the bicycle and applies the brakes as soon as possible. The reaction time of the cyclist is 0.2 s. The bicycle then decelerates uniformly and finally stops at $t = 3.8$ s.

(i) Complete the velocity-time graph below from $t = 2.0$ s to $t = 3.8$ s. (2 marks)





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(ii) The total mass of the cyclist and the bicycle is 65 kg. Calculate the resistive force for the deceleration. (2 marks)

(iii) Find the distance between the bicycle and the obstacle when the bicycle just stops. (3 marks)

(d) Figure 3.2 shows a knee pad with a padded cushion on the front to protect the cyclist in the event the knee hits the ground. Explain the working principle of this padded cushion. (2 marks)

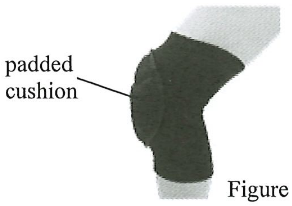


Figure 3.2

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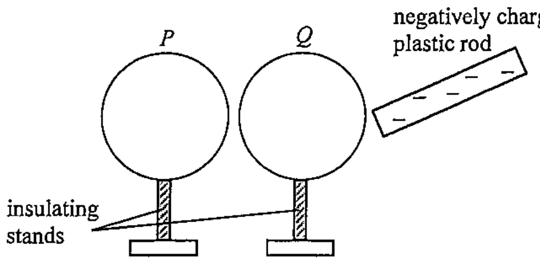
9. Figure 9.1 shows two identical conducting light spheres $P$ and $Q$ placed close to each other without touching.

Figure 9.1




Both spheres are uncharged initially. A negatively charged plastic rod is brought close to $Q$ but without touching the sphere.

(a) On Figure 9.1, sketch the distribution of charges on $P$ and on $Q$. (2 marks)

A student touches sphere $P$ momentarily with his finger while the plastic rod is in the position as shown. The plastic rod is removed afterwards.

(b) Explain whether the amount of charges on the plastic rod would decrease. (2 marks)

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(c) **The two light spheres** are now suspended by nylon threads from fixed supports such that the spheres are far apart from each other. Explain how you would use one of the spheres to test for the unknown polarity of another charged rod. (2 marks)


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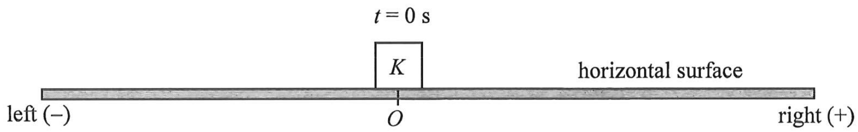
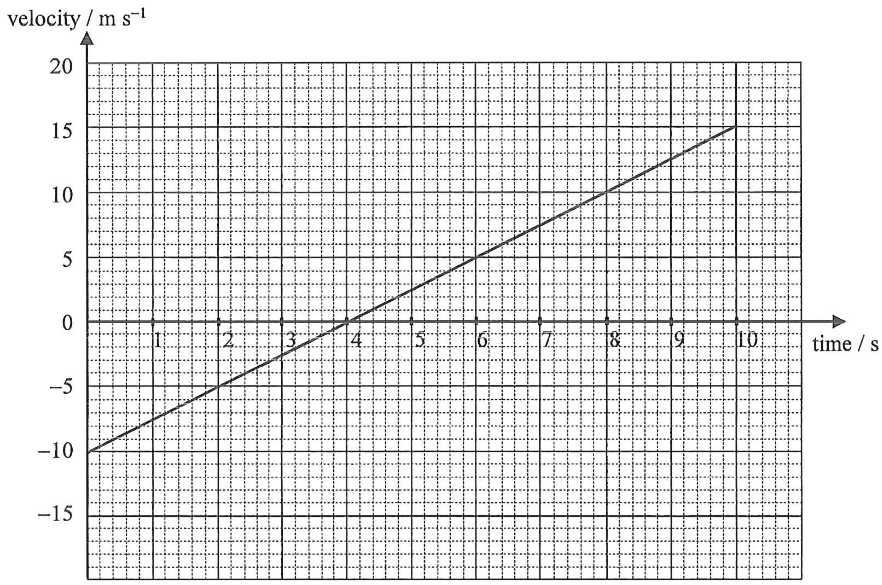
3.

Figure 3.1




An object $K$ moving on a horizontal surface is acted upon by a horizontal constant net force (not drawn on the diagram) such that it passes point $O$ at time $t = 0$ s as shown in Figure 3.1. Its velocity-time graph is shown below, taking to the right as positive (+).

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(a) Briefly describe the motion of the object from $t = 0$ s to $t = 4$ s. (2 marks)

(b) Find the position of the object at $t = 10$ s. (2 marks)

(c) Find the magnitude of the net force acting on the object at $t = 4$ s. The mass of the object is 0.4 kg. (2 marks)



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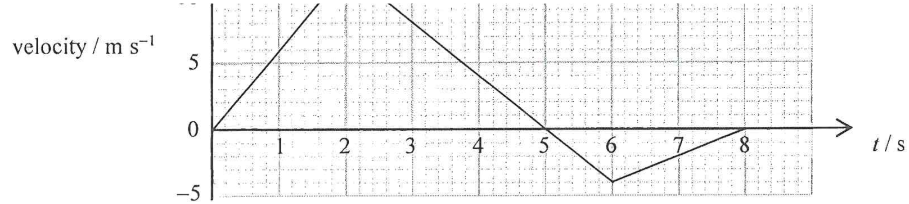
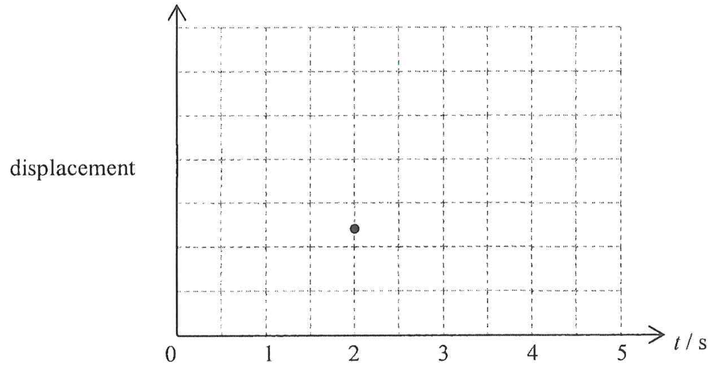



(i) Find the maximum height that the block can reach from $t = 0$ to 8 s. (2 marks)

(ii) Sketch the displacement-time graph of the block from $t = 0$ to 5 s. The displacement at $t = 2$ s is marked. Labelling of values is not required and the displacement at $t = 0$ s is taken to be zero. (2 marks)




(iii) Find the tension of the string at $t = 5$ s. (3 marks)

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(i) Draw an arrow in the figure below to represent the direction of the acceleration of the block at that moment. (1 mark)

(ii) Find the horizontal displacement of the block from that moment until just before it reaches the ground. (3 marks)

(iii) Describe the energy conversion of the block from that moment until just before it reaches the ground. (2 marks)

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4. One end of a piece of string is fixed to a wall. A wave pulse travels along the string at a speed of $0.5 \text{ m s}^{-1}$ towards the fixed end. Figure 4.1 shows the string at time $t = 0$ s and $t = 2$ s.

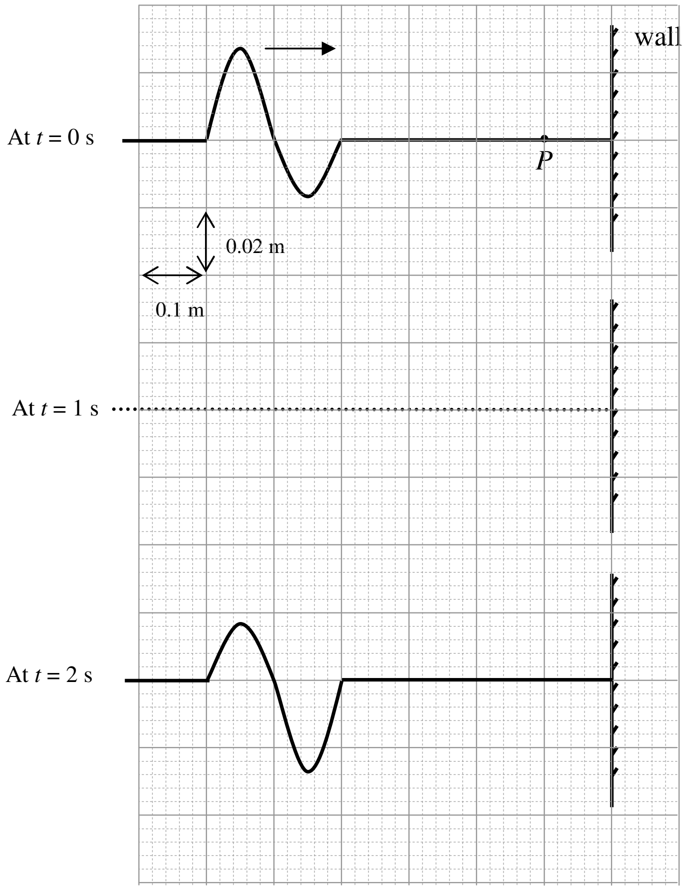
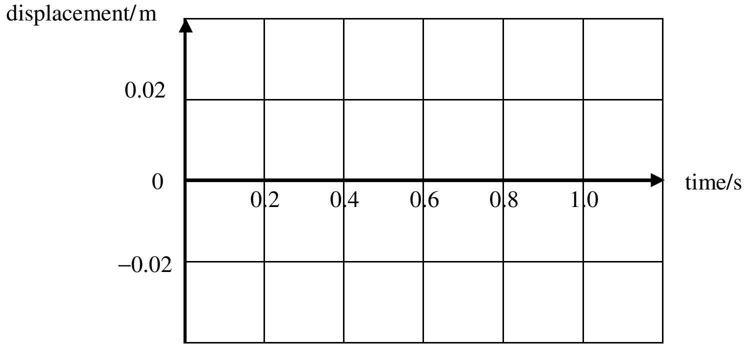



**Figure 4.1**

(a) On Figure 4.1, draw the shape of the wave pulse at t = 1 s. (1 mark)

(b) Sketch a graph of the displacement of point $P$ on the string at a distance of 0.1 m from the wall during the period $t = 0$ s to $t = 1$ s. (2 marks)

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9.

**Figure 9.1**

Figure 9.1 shows a microwave oven. Mary wants to estimate the useful output power of the oven. She is provided with the apparatus and material shown in Figure 9.2.

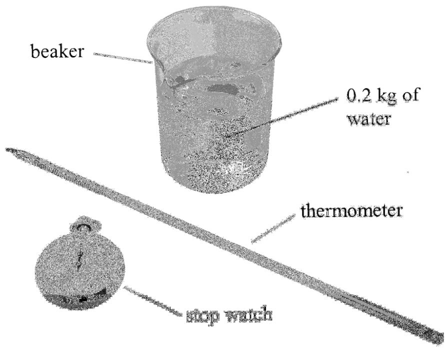


**Figure 9.2**

(a) Describe how Mary should conduct the experiment. Specify all measurements Mary has to take. State **EITHER** one precaution taken **OR** one assumption made when conducting this experiment. Write down an equation for calculating the useful output power. (5 marks)



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(b) The value obtained by Mary is found to be smaller than the rated power of the oven. Suggest one possible reason to account for this difference. (1 mark)

(c) Explain whether increasing the mass of water used in the experiment would improve the accuracy of the experiment. (1 mark)

10.

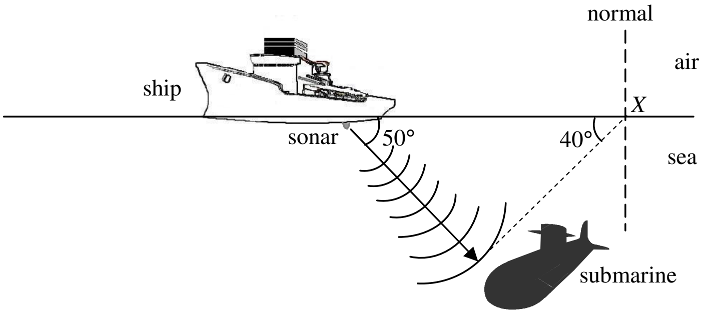


**Figure 10.1**

Figure 10.1 shows a ship equipped with sonar. The sonar emits ultrasonic waves of frequency 25 kHz into the sea. The waves propagate at an angle of 50º to the surface of the sea and are reflected from a submarine back to the ship after 0.15 s.

Given : speed of sound in air = 340 m $s^{-1}$
speed of sound in sea water = 1500 m $s^{-1}$

(a) Calculate the vertical distance of the submarine beneath the sea surface. (2 marks)

(b) Some of the reflected waves propagate along the dotted line and emerge into the air at *X*. Calculate the angle of refraction in air. (2 marks)

(c) Is it possible for ultrasonic waves, at certain angles of incidence, to undergo total internal reflection when they go from sea water to the air ? Explain. (2 marks)

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